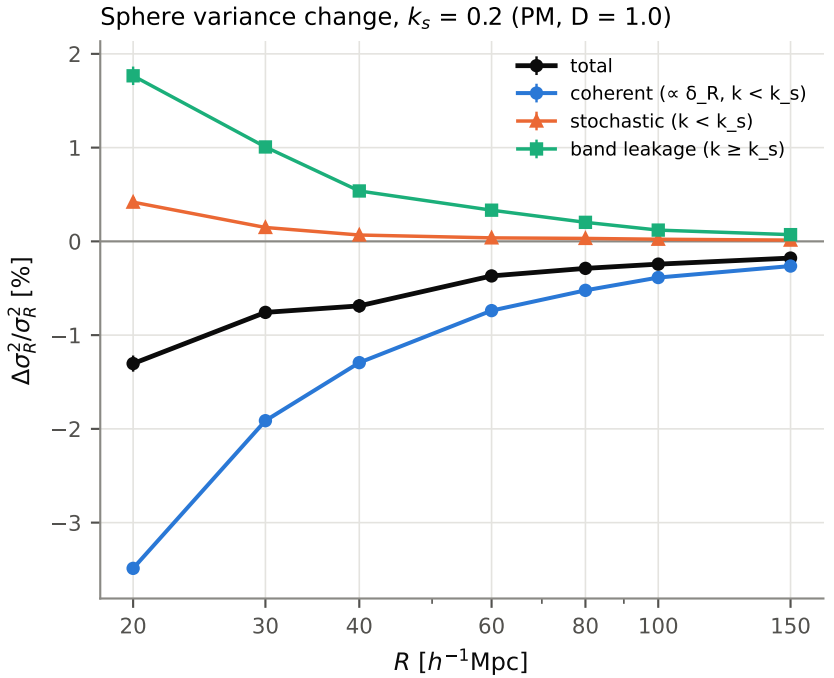
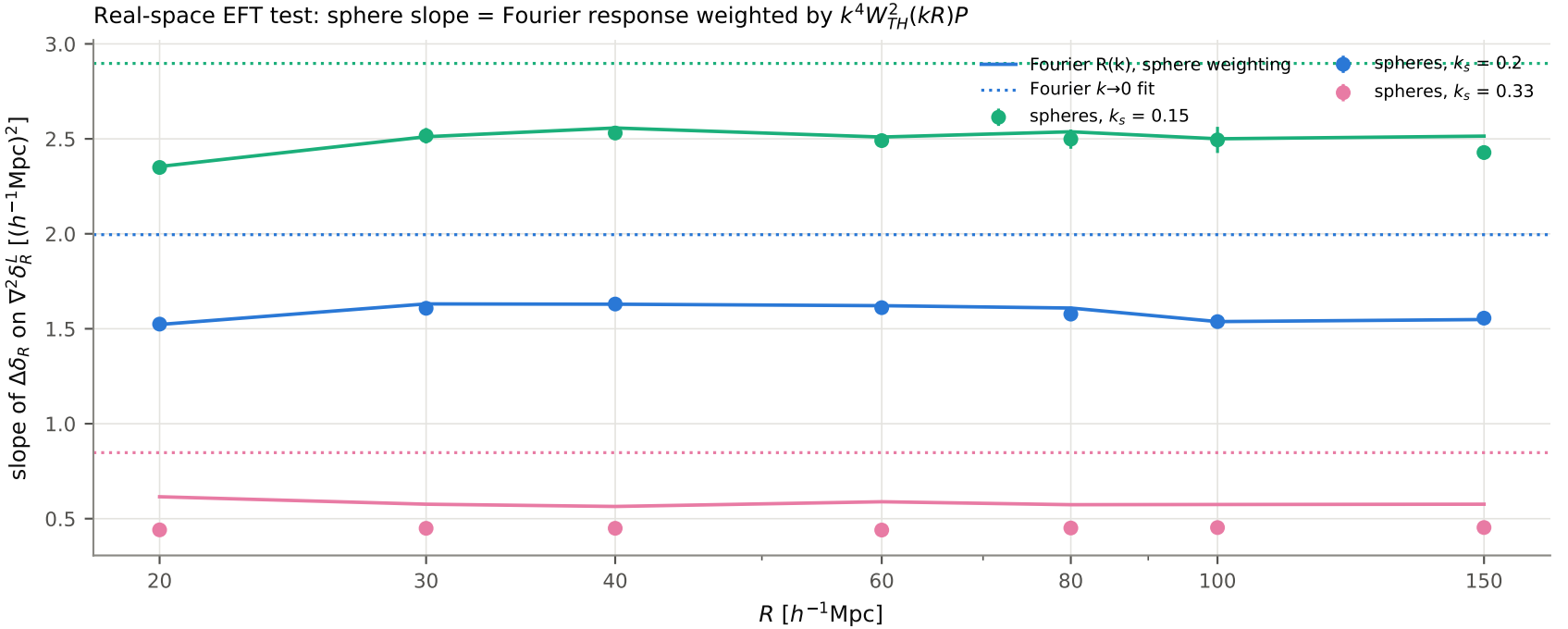
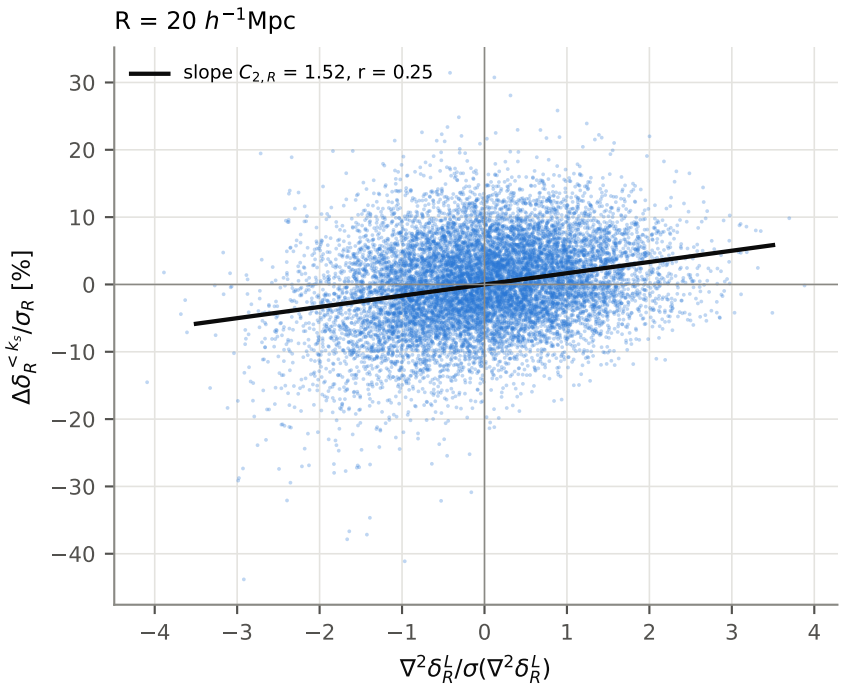
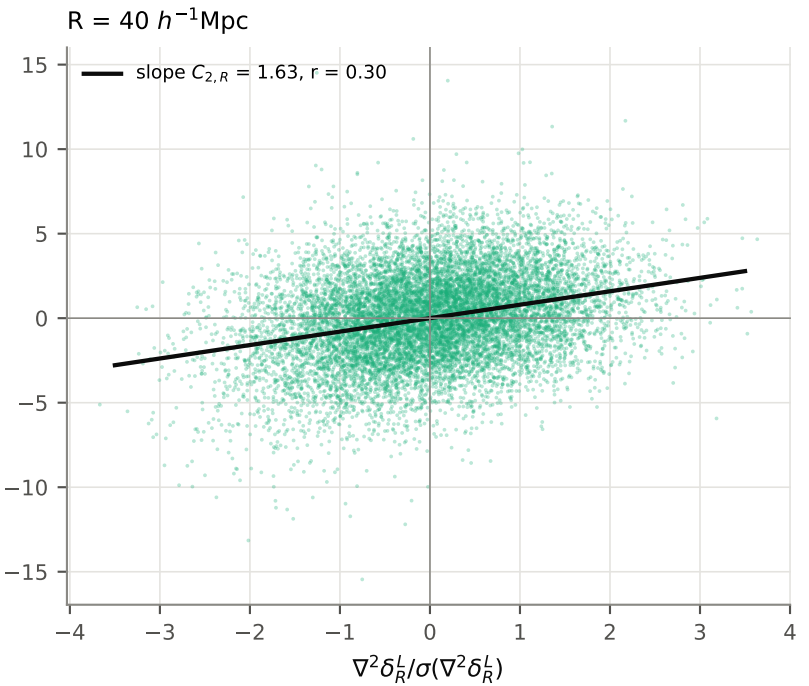
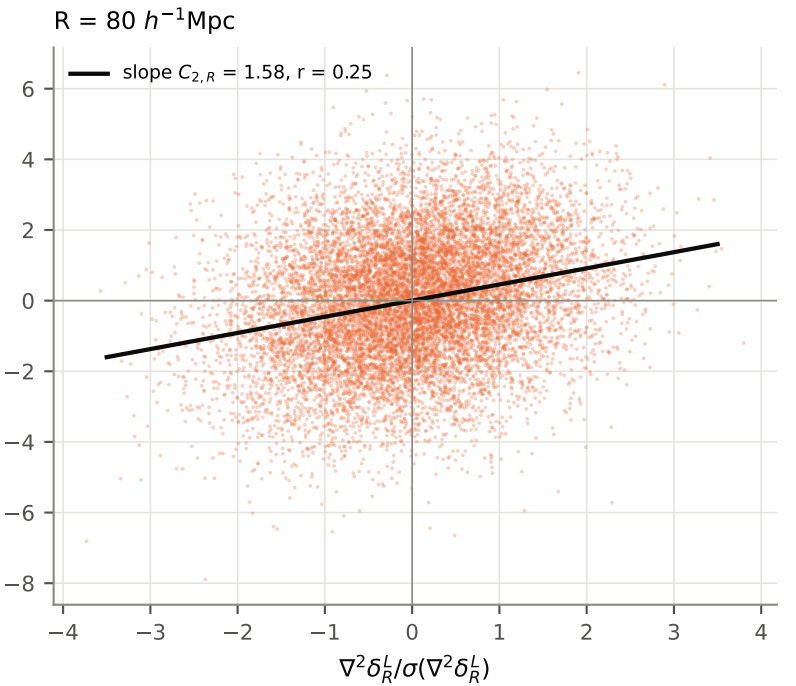

Sphere variance change, k_s = 0.2 (PM, D = 1.0)
total
coherent (∝ δ_R, k < k_s)
stochastic (k < k_s)
band leakage (k ≥ k_s)
$\Delta\sigma_R^2/\sigma_R^2$ [%]
R [h^{-1}Mpc]


Real-space EFT test: sphere slope = Fourier response weighted by $k^4 W_{TH}^2(kR)P$
Fourier R(k), sphere weighting
Fourier $k\to0$ fit
spheres, k_s = 0.15
spheres, k_s = 0.2
spheres, k_s = 0.33
slope of $\Delta\delta_R$ on $\nabla^2\delta_R^L$ [$(h^{-1}\mathrm{Mpc})^2$]
R [h^{-1}Mpc]


R = 20 h^{-1}Mpc
slope $C_{2,R}$ = 1.52, r = 0.25
$\Delta\delta_R^{<k_s}/\sigma_R$ [%]
$\nabla^2\delta_R^L/\sigma(\nabla^2\delta_R^L)$


R = 40 h^{-1}Mpc
slope $C_{2,R}$ = 1.63, r = 0.30
$\nabla^2\delta_R^L/\sigma(\nabla^2\delta_R^L)$


R = 80 h^{-1}Mpc
slope $C_{2,R}$ = 1.58, r = 0.25
$\nabla^2\delta_R^L/\sigma(\nabla^2\delta_R^L)$